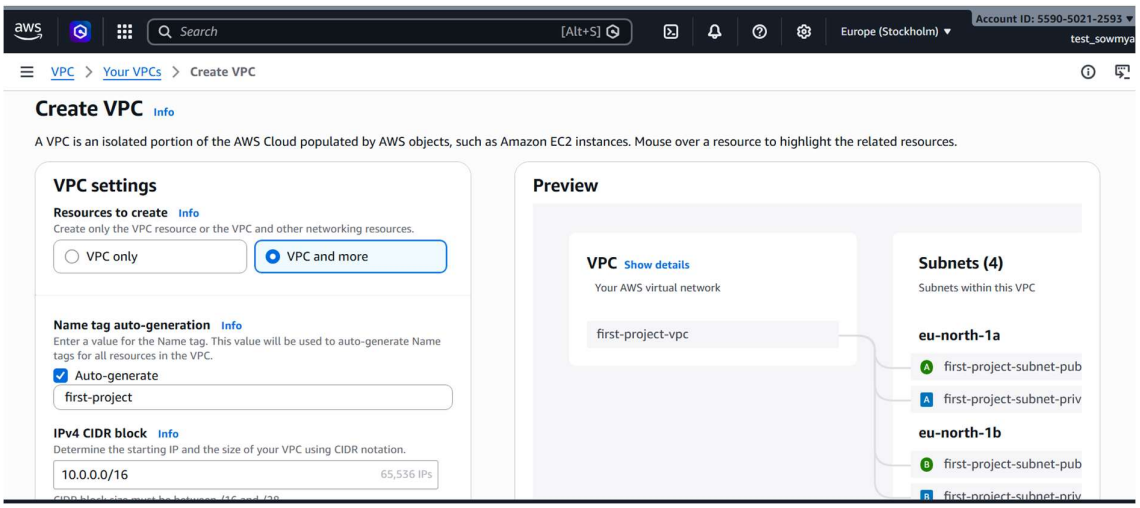
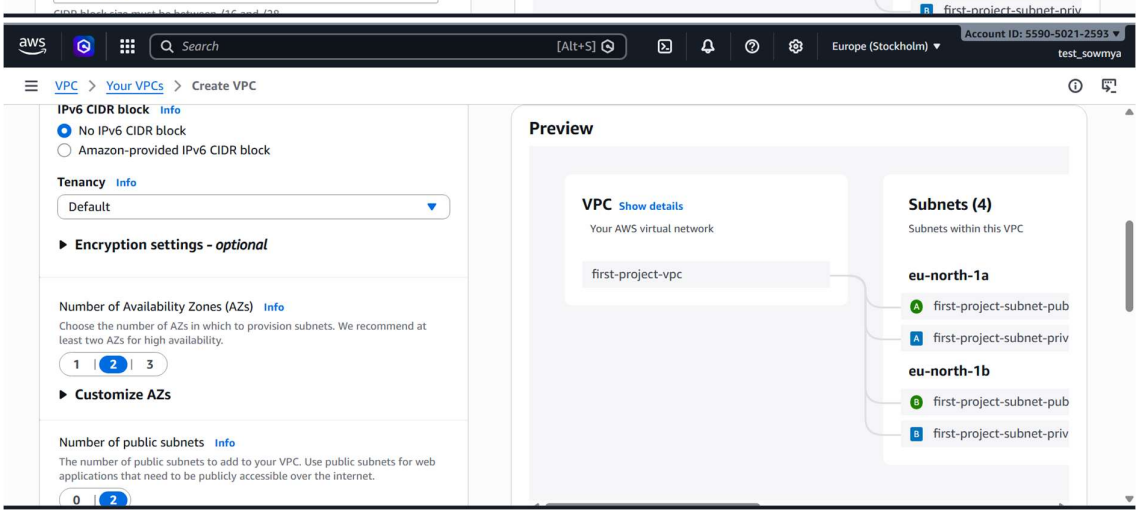
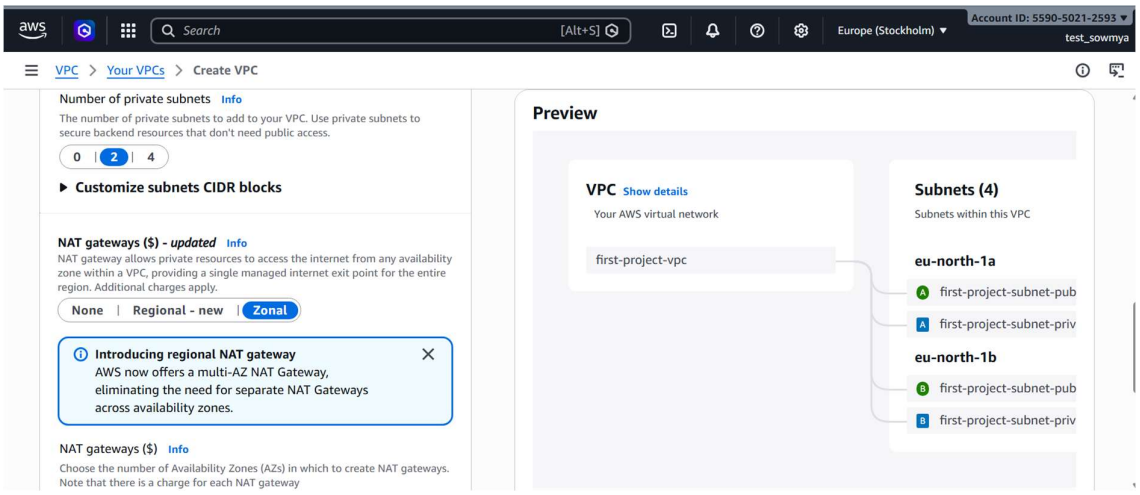
Project: AWS Production Architecture Project-VPC, Bastion Host, NAT Gateway, ELB and EC2.

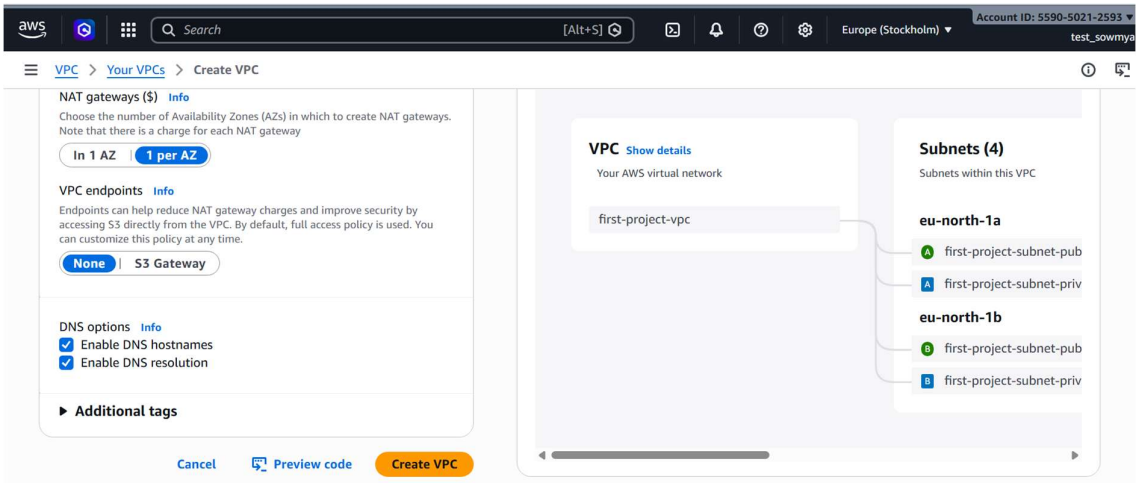
Step 1 : Login to your AWS console with your IAM user and type VPC in search bar

Step 2: Click on create VPC and select VPC and more so that you can create subnets, route tables etc.. in one go.

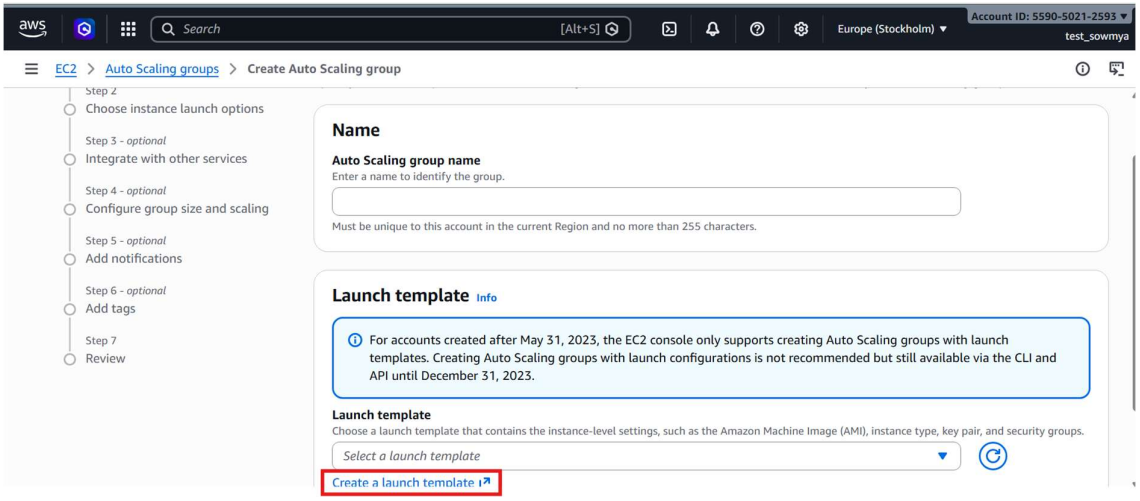








Step 3: In search bar type EC2 and in the bottom left corner you have Auto scaling group, Select it and create a Auto scaling group(ASG). Here before creating ASG you have to create a launch template for ASG. Click on create launch template



Follow steps as shown in below images for creating launch template.

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

first-project

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

Application deployment in AWS private subnet

Max 255 chars

Auto Scaling guidance Info

Select this if you intend to use this template with EC2 Auto Scaling

☑ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Create launch template

---

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Launch templates > Create launch template

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type — Free tier eligible
ami-080254318c2d8932f (64-bit (x86)) / ami-0af742bfb3356fe77 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Publish Date	Username
64-bit (...	ami-080254318c2d8932f	2026-03-14	ubuntu Verified provider

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

Virtual server type (instance type)
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New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Create launch template

---

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Launch templates > Create launch template

Instance type

t3.micro — Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour
On-Demand SUSE base pricing: 0.0108 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

first-project | Create new key pair

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

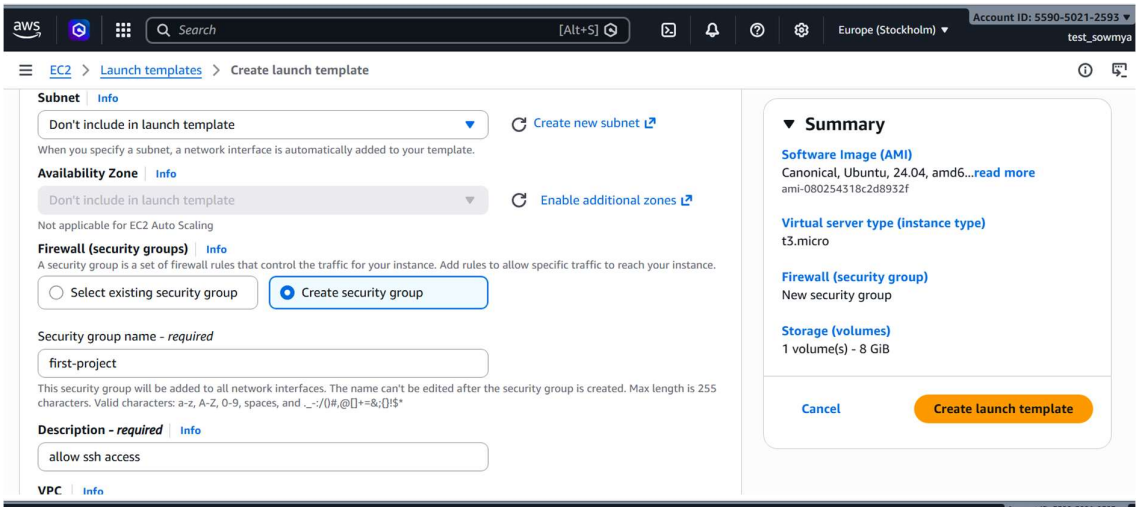
Virtual server type (instance type)
t3.micro

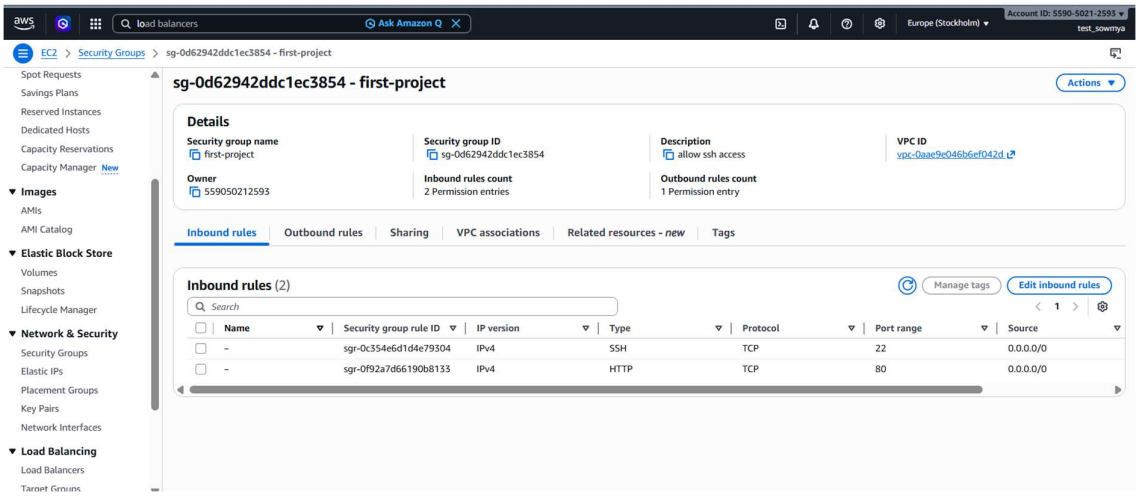
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Create launch template

Network settings Info

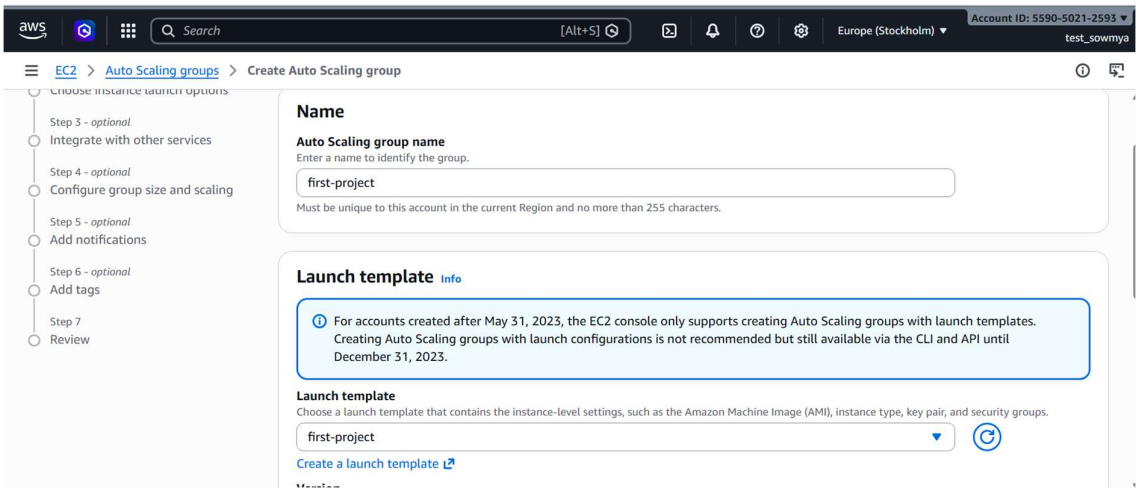


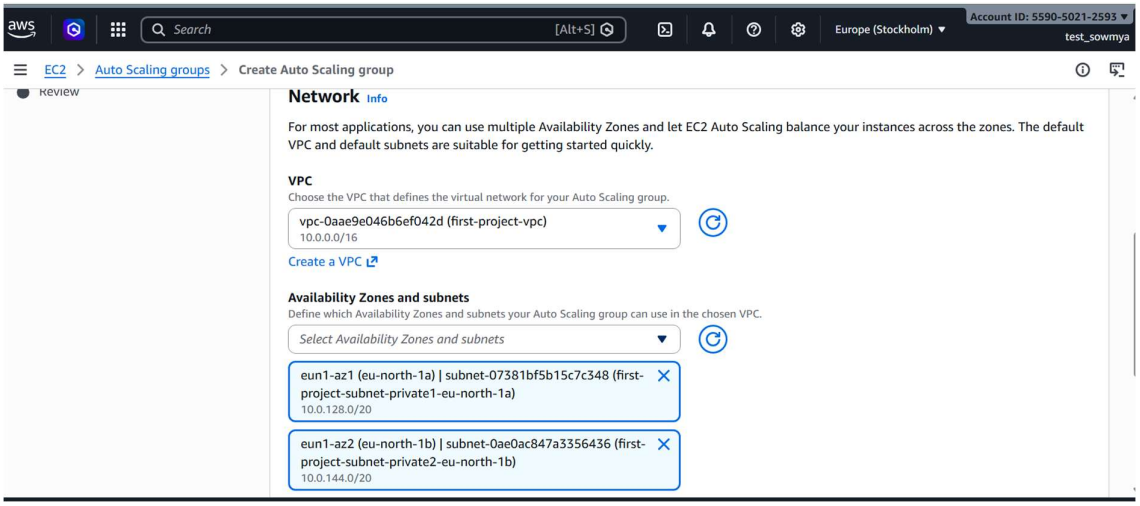


Here select the port as per your application. I'm running apacher server so I selected port 80.

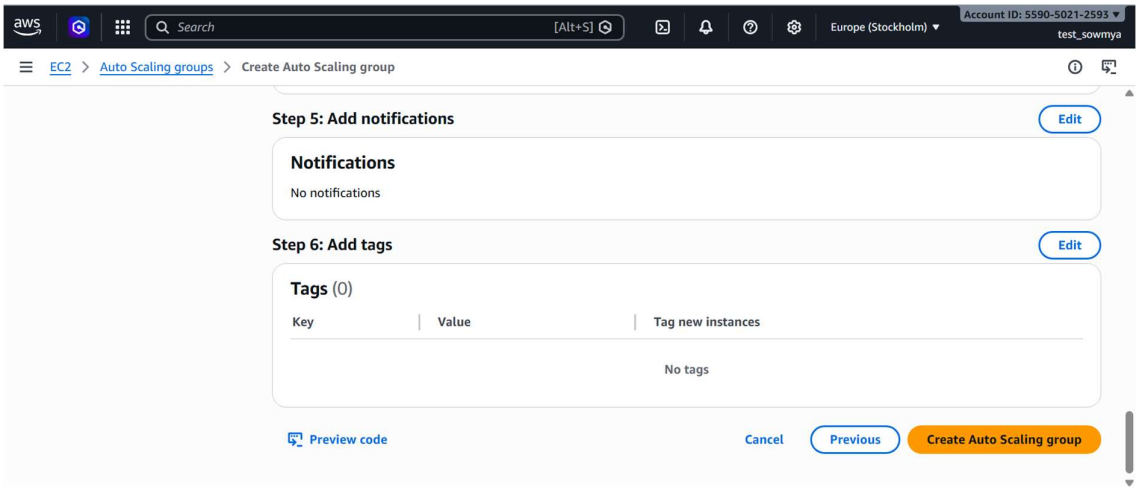
Now click on create launch template.

Step 4: Now go back to ASG and refresh the page, Now create ASG. Follow the steps as shown in below images





Click on next button and review everything once and click on create ASG.



Step 5: Now will deploy application on EC2 instances in private subnets. Before that we need to create a bastion host instance to access our application(as the private subnets do not have public IP)

Go to EC2 and launch a new instance.

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Instances > Launch an instance

Name and tags Info

Name

bastion-host

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux

Browse more AMIs

Including AMIs from

Summary

Number of instances Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel | Launch instance | Preview code

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Instances > Launch an instance

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type — Free tier eligible
ami-080254318c2d8932f (64-bit (x86)) / ami-0af742bfb3356fe77 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Publish Date	Username	
64-bit (...	ami-080254318c2d8932f	2026-03-14	ubuntu	Verified provider

Instance type Info | Get advice

Instance type

Summary

Number of instances Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Cancel | Launch instance | Preview code

aws | Search [Alt+S] | Europe (Stockholm) | Account ID: 5590-5021-2593 | test_sowmya

EC2 > Instances > Launch an instance

Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour
On-Demand SUSE base pricing: 0.0108 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

first-project

Create new key pair

Summary

Number of instances Info

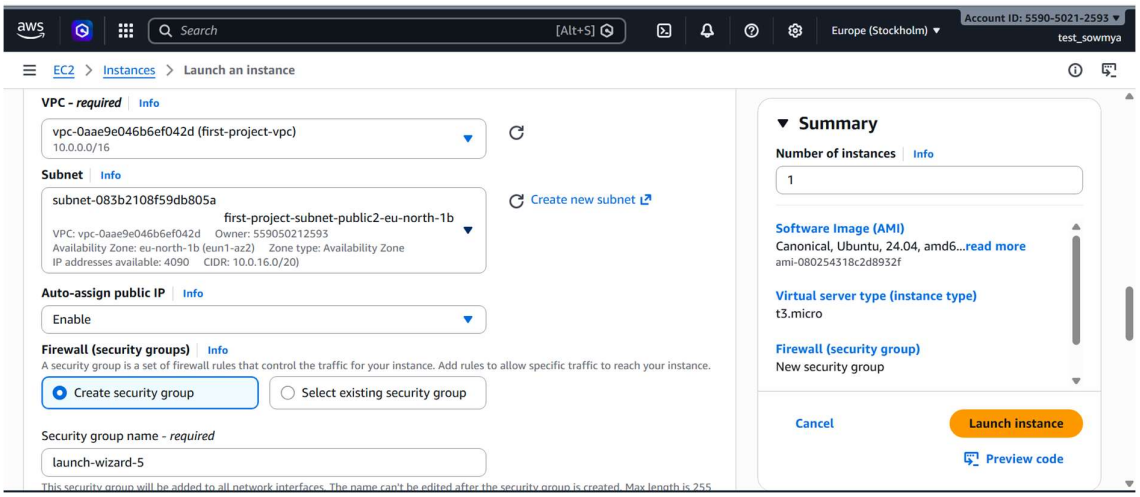
1

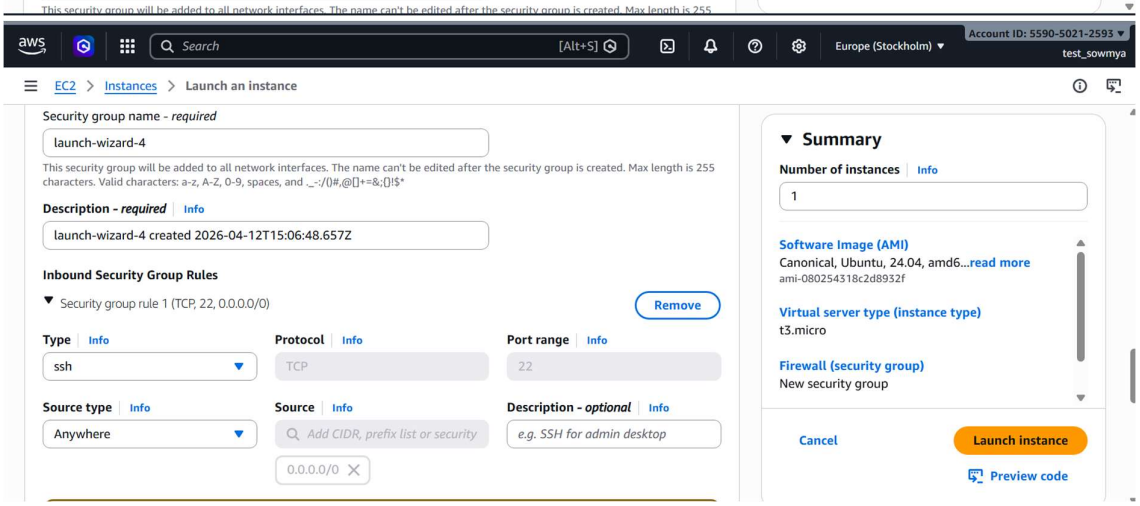
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-080254318c2d8932f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

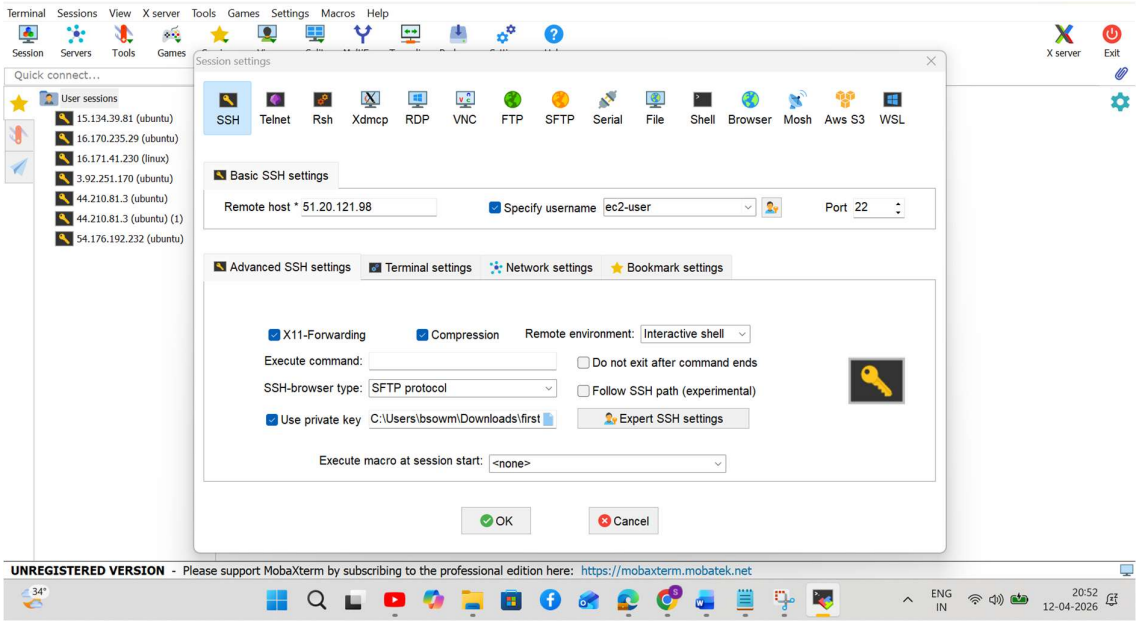
Cancel | Launch instance | Preview code





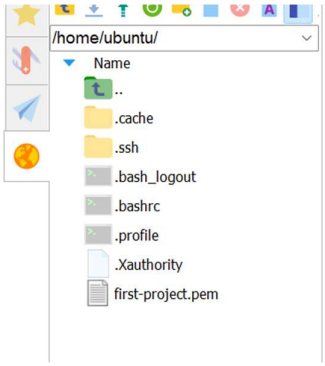
To SSH to the private subnet we need key value pair login so first we copy the key value pair credentials to bastion host(as key value pair is present on the laptop).For Bastion host to login to private subnet the key value pair is required.

Step 6: Here I'm using MobaXterm to connect to EC2 instances. Open MobaXterm select Session->SSH->paste the public ip of bastion instance->advanced SSH settings->select private key and select the private key location.Click on OK



If this fails go to local terminal in the MobaXterm then run following commands

ssh -i path of your pem file ubuntu@16.171.0.98 or simply drag the .pem file and put it in leftside of mobaxterm terminal as shown in below image



Then give permission to the file with below command

chmod 400 first-project.pem

Then finally use ssh -i first-project.pem ubuntu@<PRIVATE-IP-OF-YOUR-PRIVATE-EC2>.

Now you will be inside the private subnet in mobaxterm.

Step 7: Now install the webserver using below commands in mobaxterm (here I am installing apache server)

sudo apt update -y

sudo apt install apache2 -y

sudo systemctl start apache2

sudo systemctl enable apache2

Step 8: Installation completed. Now I will create a html file so that I can access it in the server

Commands

cd /var/www/html (Apache serves web pages from this directory**)**

sudo vim first.html

press i and insert following code

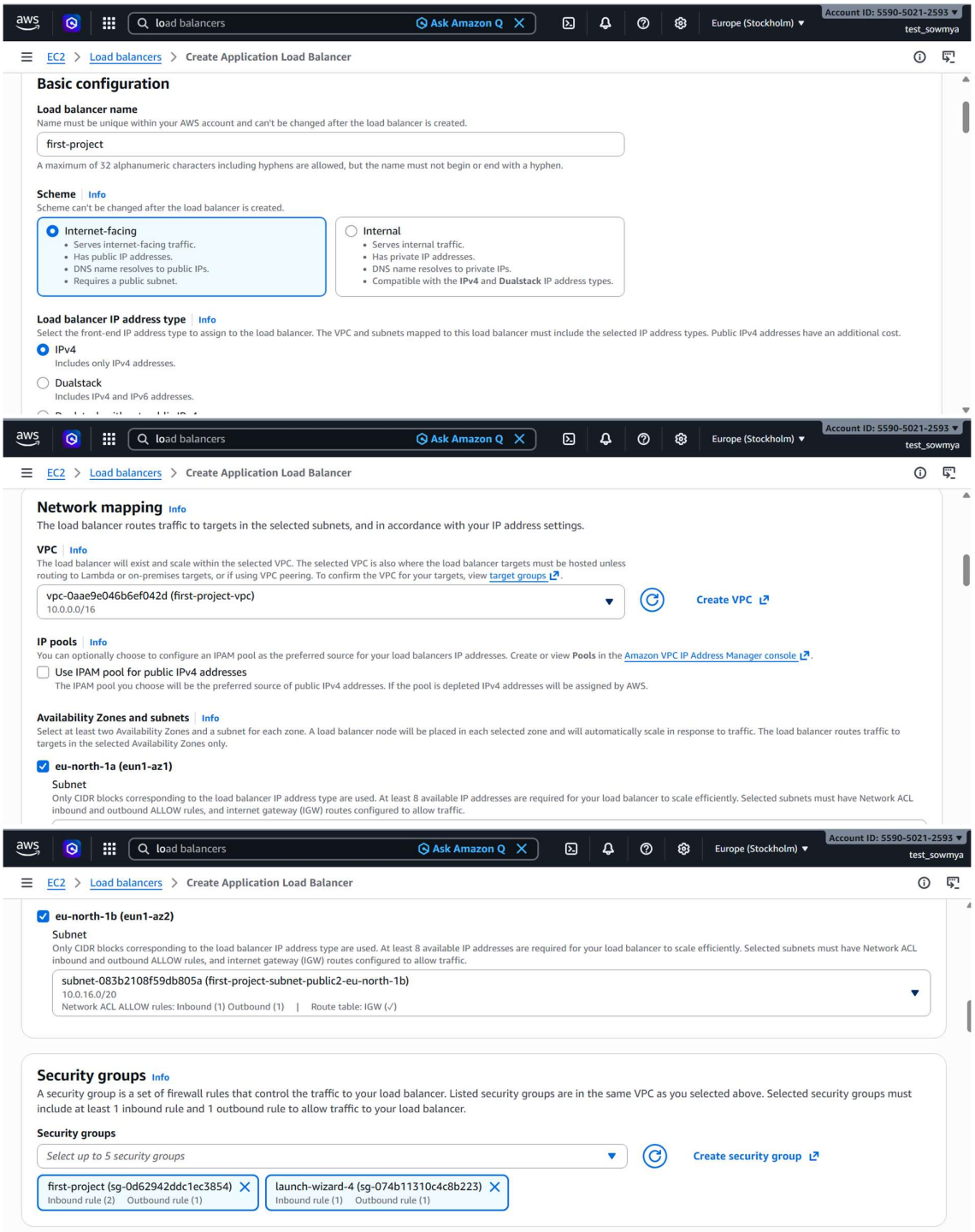
```
<html>
  <head>
    <title>My AWS Project</title>
  </head>
  <body>
    <h1>Hello from Private EC2 </h1>
    <p>This is served using Apache inside private subnet</p>
  </body>
</html>
```

Then press ESC and save using :wq!

To verify u can use cat first.html

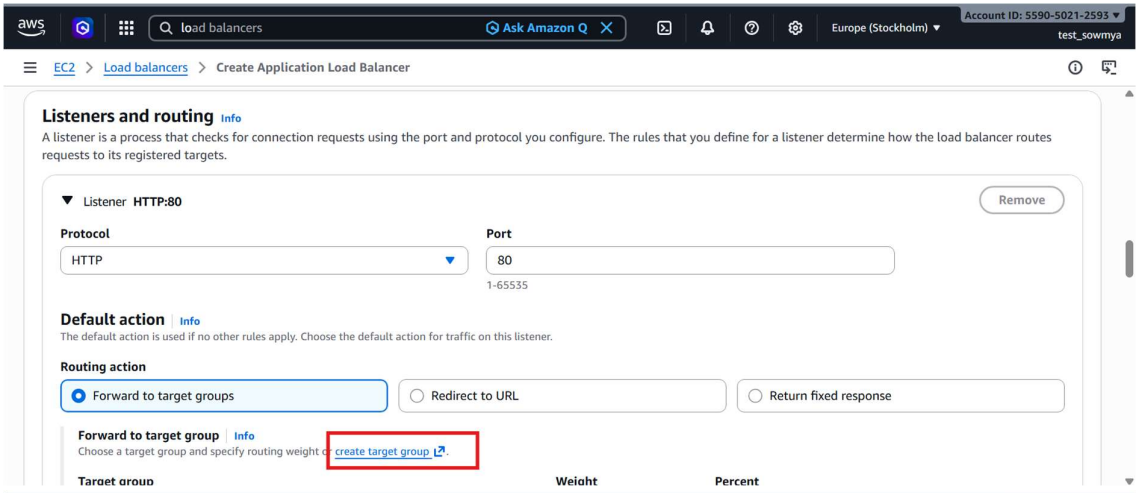
Remember here I have just logged into one of the private instance. So if u try to access the other private subnet it will give you error as the application is only available in 1 instance. If you want you can repeat the same steps with other private instance too.

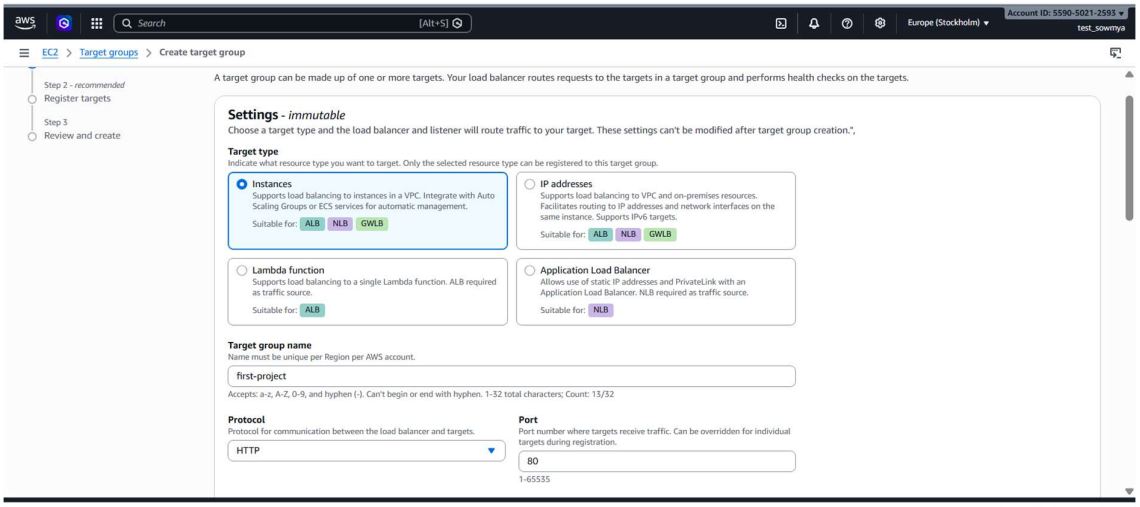
Step 9 : Now Create a Application load balancer.

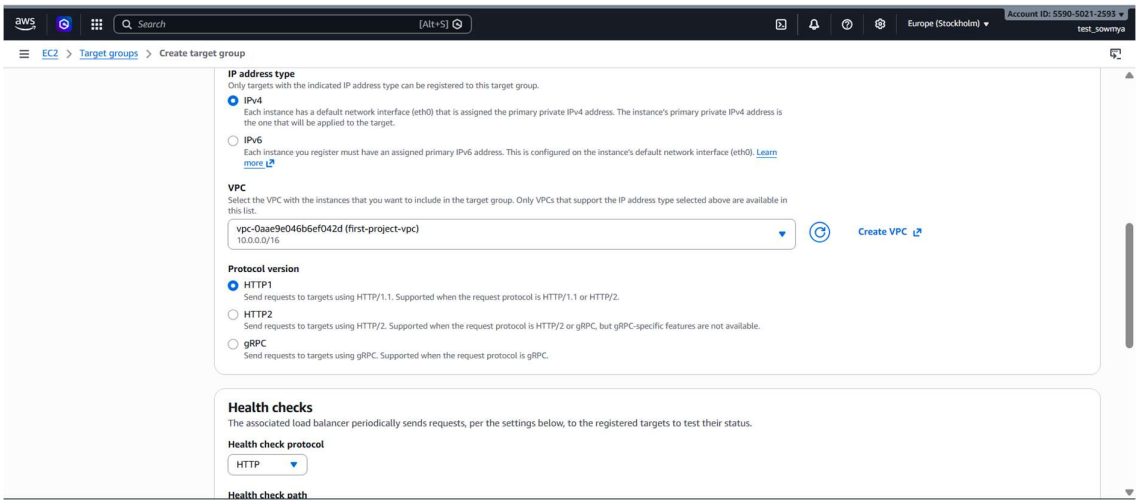


Now here click on create target group as shown in below image

We create a Target Group first because the Load Balancer needs to know where to send traffic.

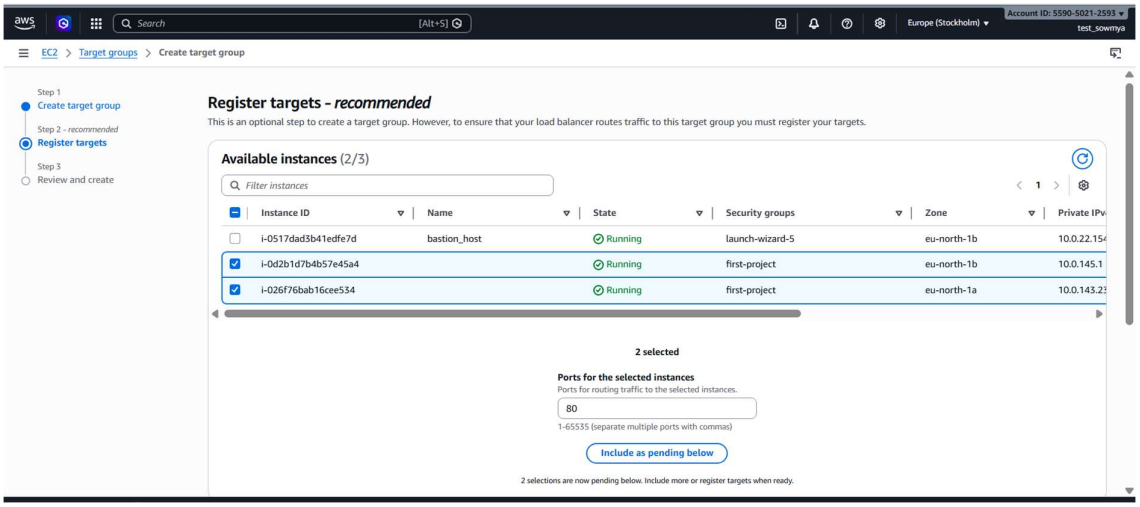


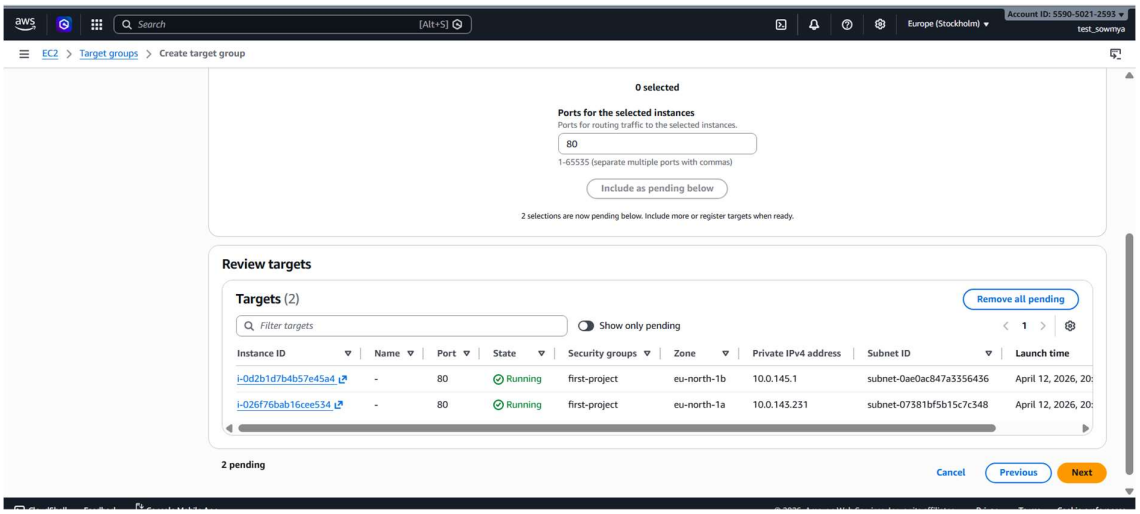




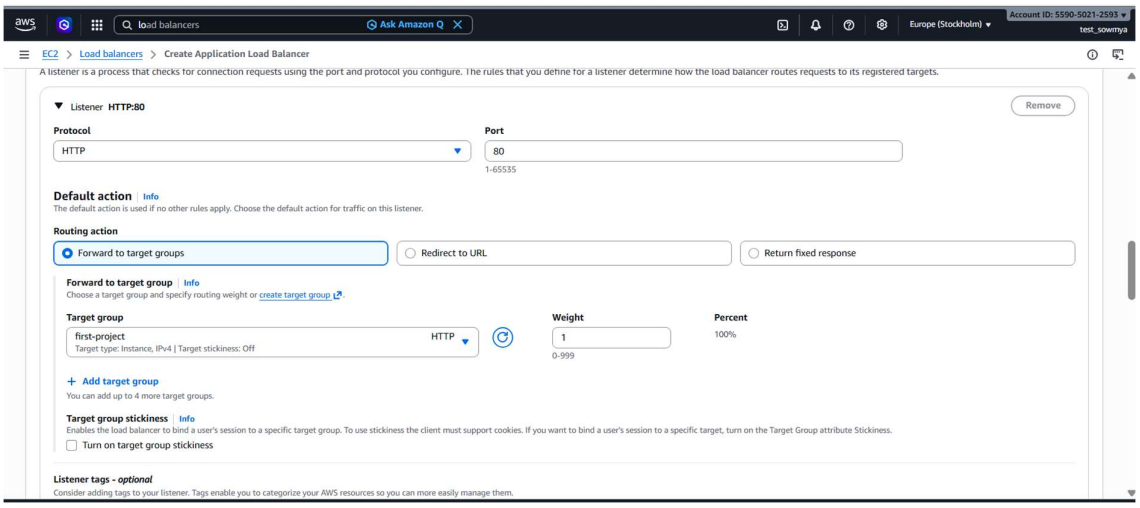
Then click on next . Select both private instances and register them as targets

then click on next and click on create target groups



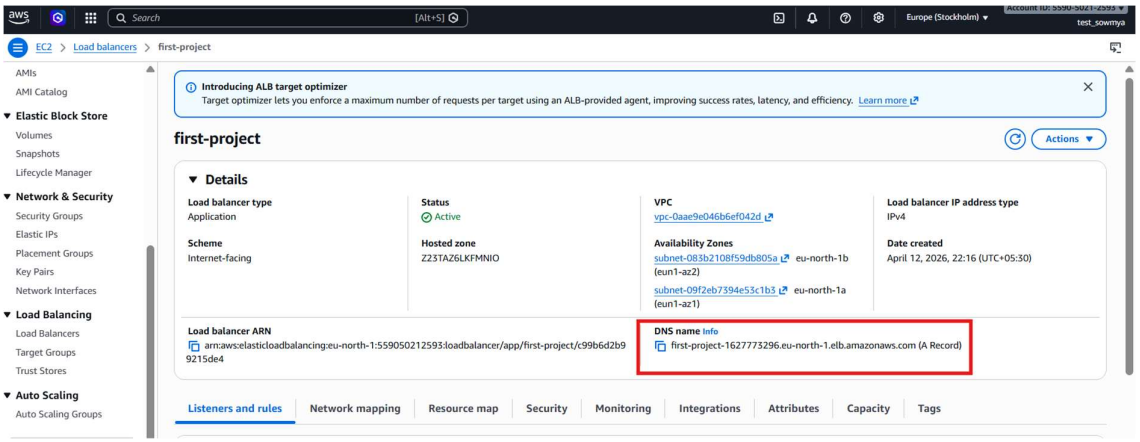


Now select the target group you have created in the loadbalancer as shown below



Finally click on create load balancer

Step 10: Go to load balancer and copy the dns name and paste it in new tab



When do it you will as in image below. Here you are unable to see the contents you created in first.html because Apache serves index.html as the default file when accessing the root URL.

So for this execute the following command in the mobaxterm terminal

sudo mv first.html index.html

Now if you load the page again you will able to see the contents

Hurray!!! We have successfully deployed our application.